PAPER_319: Compact HII SFR Gravitational Binding Phase Transition

Author: Daniel T. Murphy **Date:** March 2026 ## t_cross = 67,730 yr | sSFR = 5×10^{-4} yr^{-1} (50× Lagoon) | m_factor(t_age) = 151 ### FIRST UQFF Compact HII Region SFR Runaway Gravitational Binding Phase Transition

Session: 91
Module: ORION_UQFF_MODULE.cpp (33rd C++ module)
System: Orion Nebula M42/NGC 1976 — compact HII region with active star formation
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Abstract

This paper derives the SFR-driven gravitational binding phase transition for the Orion Nebula. With SFR = 1 M_sun/yr acting on an initial mass M = 2000 M_sun, the specific star formation rate **sSFR = 5×10^{-4} yr^{-1} is 50× that of the Lagoon Nebula** (1×10^{-5} yr^{-1}, PAPER_305). The system is born wind-dominated (unbound), but as SFR continuously adds mass, the effective gravitational acceleration amplified by the SFR mass factor m_factor(t) = 1 + SFR×t_yr/M_sun_count crosses the growing wind ram pressure at **t_cross = 67,730 yr**, transitioning the nebula from unbound to gravitationally bound. By t_age = 300 kyr, binding_ratio = g_SFR/a_wind = 2.654. This is the FIRST UQFF compact HII SFR runaway gravitational binding phase transition.

System Parameters

Parameter	Value	Description
M	2000 M_sun = 3.978×10^{33} kg	Initial mass
SFR	1 M_sun/yr = 6.303×10^{22} kg/s	Star formation rate
sSFR	5×10^{-4} yr^{-1}	Specific SFR = SFR/M
t_age	3×10^{5} yr	Nebula age
v_wind	8×10^{3} m/s	HII ionization front
g_base	1.907×10^{-11} m/s^2	Base gravity (PAPER_317)
a_wind(t=0)	5.424×10^{-10} m/s^2	Initial wind acceleration

Key Equations

SFR Mass Factor

$$M_{\text{sf}}(t) = \frac{\text{SFR}_{\text{yr}} \times \text{t}_{\text{yr}}}{M_{\text{sun,count}}}, \qquad m_{\text{factor}}(t) = 1 + M_{\text{sf}}(t)$$

SFR-Amplified Gravitational Acceleration

$$g_{\text{SFR}}(t) = g_{\text{base}} \times m_{\text{factor}}(t) = g_{\text{base}} \left(1 + \frac{\text{SFR}_{\text{yr}} \cdot t_{\text{yr}}}{M_{\text{sun,count}}}\right)$$

Wind Acceleration (time-evolving)

$$a_{\text{wind}}(t) = \frac{v_{\text{wind}}^2}{r}\left(1 + \frac{t}{t_{\text{age}}}\right)$$

Crossover Time t_cross (PAPER_319 Key Result)

Setting g_SFR(t) = a_wind(t):

$$g_{\text{base}}\,(1 + \text{sSFR}\cdot t) = a_{\text{wind},0}\left(1 + \frac{t}{t_{\text{age,yr}}}\right)$$

$$t_{\text{cross}} = \frac{a_{\text{wind},0} - g_{\text{base}}}{g_{\text{base}}\cdot \text{sSFR} - a_{\text{wind},0}/t_{\text{age,yr}}} = \boxed{67{,}730\ \text{yr}}$$

Specific SFR

$$\text{sSFR} = \frac{\text{SFR}_{\text{yr}}}{M_{\text{sun,count}}} = \frac{1}{2000} = 5\times 10^{-4}\ \text{yr}^{-1} = 50\times \text{Lagoon}$$

Gas Depletion Timescale

$$t_{\text{consume}} = \frac{M_{\text{sun,count}}}{\text{SFR}_{\text{yr}}} = \frac{2000}{1} = 2000\ \text{yr} \quad \text{(without OMC-1 replenishment)}$$

Results Summary

Quantity	Value	Significance
sSFR	**5×10^{-4} yr^{-1}**	50× Lagoon Nebula
t_cross	**67,730 yr**	Unbound → bound transition
m_factor(t_age=300 kyr)	**151**	SFR amplification
g_SFR(t_age)	2.878×10^{-9} m/s^2	151× g_base
a_wind(t_age)	1.085×10^{-9} m/s^2	
binding_ratio(t_age)	**2.654**	Gravitationally bound
m_factor(1 Myr)	**501**	
binding_ratio(1 Myr)	**4.069**	Increasingly bound
t_consume	**2000 yr**	Gas depletion (no replenishment)

Phase Transition Diagram

```
t=0          t_cross=67.7 kyr    t_age=300 kyr     t=1 Myr
|                |                    |                 |
UNBOUND          TRANSITION           BOUND 2.65×       BOUND 4.07×
a_wind>g_SFR     a_wind = g_SFR       g_SFR>a_wind      g_SFR>>a_wind
η=28.47 wind     ─── crossover ───    η_B=2.654         η_B=4.069
```

Comparison: UQFF SFR Class

Module	SFR (M_sun/yr)	M (M_sun)	sSFR (yr^{-1})	PAPER
Lagoon	0.1	10,000	1×10^{-5}	PAPER_305
Orion	**1.0**	**2000**	$\mathbf{5\times10^{-4}}$	**PAPER_319**
M16 (Eagle)	~0.01×	1200	—	PAPER_284

Orion sSFR is 50× Lagoon — UQFF identifies this as the "ultra-compact HII" class with rapid gravitational binding crossover. Lagoon has t_consume = 100 kyr; Orion has t_consume = **2000 yr**, the shortest gas depletion time in the UQFF series, sustained only by continuous OMC-1 giant molecular cloud inflow.

UQFF Physical Interpretation

The phase transition at t_cross = 67,730 yr is a structural boundary in Orion's evolution:

- **Before t_cross:** The system is wind-dominated (η_wind > 1). UV photoionization and ram pressure drive a champagne flow outward; newly formed stars are subject to wind erosion.
- **After t_cross:** SFR has added sufficient mass that g_SFR > a_wind. The system becomes self-gravitating with respect to its own stellar formation. The cluster proceeds to form stars under its own gravity — consistent with the Orion OB1 association framework where the cluster is now 300 kyr old and gravitationally bound.

Within ORION_UQFF_MODULE.cpp, the SFR mass factor m_factor(t) enters the base gravity term via `G*M*(1+M_sf(t))/r^2`, registered as `WOLFRAM_TERM_ORION_BASE` and `WOLFRAM_TERM_ORION_SFR_BINDING`.

WOLFRAM_TERM Registration

```
#define WOLFRAM_TERM_ORION_BASE(val)        (val)
// g_base = (G*M*(1+M_sf(t))/r^2)*(1+Hz*t)*(1-B/B_crit)*(1+f_TRZ)
// M_sf(t) = SFR_yr*t_yr / M_sun_count  [PAPER_319 SFR amplification]

#define WOLFRAM_TERM_ORION_SFR_BINDING(val) (val)
// wind = v_wind^2/r * (1+t/t_age)  [crosses SFR gravity at t_cross=67730 yr]
```

Series first: FIRST UQFF compact HII SFR runaway gravitational binding phase transition. Establishes sSFR as a new UQFF classification axis for HII regions: compact (Orion, sSFR=5×10^{-4} yr^{-1}) vs extended (Lagoon, sSFR=1×10^{-5} yr^{-1}).

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.157 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\text{DVP}} = 71, \quad n_{\text{channel}} = 8/26$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\mathrm{DVP}} = 71$ is **resonant** (threshold at $p > 26$), the system’s vacuum topology inherits its resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^4 yr** (spin-down equilibrium): $\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

| Framework | Canonical Value | This Paper | Status | |——-|——-——-|——-|——-| | VDS ratio | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.157 | ✓ Threshold-consistent | | DVP prime | $p_k \in$ {2,3,...,113} | $p_{\text{DVP}} = 711$ | ✓ Res-

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	✓ Consistent
Cosmological constant Λ	1.1×10^{-52} m^{-2} (UQFF vacuum term)	1.114×10^{-52} m^{-2}	Planck 2018	✓ Consistent
Proton decay rate	κ = 0.0005/day $\rightarrow$ Γ_p suppression	< 4.17×10^{-35}/yr	Super-K 2024	✓ Consistent
UQFF buoyancy signature	F_U_Bi_i unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`) for full UQFF–SM bridge.